CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. Examination (EE)

November 2013

EE 303.01 Microcontroller and Applications

Time: 10.00am to 01.00pm **Date: 22-11-2013** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.
5. **Comments are mandatory with each line of code.**

SECTION – I

Q-1 (a) 1. What do you mean by Non Maskable & Maskable interrupts? Explain various bits of IE register. **5**

2. List out peripherals available in 8051 microcontroller. **2**

Q-2 (a) 1. Explain architecture of Von Neumann & Harvard with necessary diagram. **4**

2. Draw Pin diagram & explain function of each pin for 8051 microcontroller. **5**

3. Draw internal Memory (RAM & ROM) Architecture for 8051 microcontroller with information of address range for each segment of memory. **5**

OR

(a) 1. Compare Microprocessor & Microcontroller. **4**

2. Draw the internal Structure of PORT 0 & PORT 2 ,clearly indicating difference between them. **5**

3. For 12 MHz crystal frequency calculate time of execution for following code in 6 clock mode. **5**

Label	Instruction	Machine cycle
	MOV R3,#200	1
Here:	DJNZ R3,Here	2
	RET	2

Q-3 (a) 1. Initialize an array of 8 elements of unsigned char type. Send it to Port1 one by one. **4**

2. Explain various logical operators used for logical operations. **5**

3. Write a program to simulate infinite loop using 'while'& 'for' control loops. **5**

OR

(a) 1. Demonstrate use of various control loops (Atleast 4) in Embedded C programming. 4

2. Explain various arithmetic operators used for arithmetic operations. 5

3. Write & explain program, which separate out the two 8 bit number from a 16 bit variable named load_timer & then loads it in to TL0 & TH0. 5

SECTION – II

Q-4 (a) 1. Draw internal architecture for Timer0. 2

2. Write a program which overflows the timer at rate of 100 microseconds with 11.0592 MHz crystal oscillator frequency. Show necessary calculations for the same. 5

Q-5 (a) 1. Explain the Polling method & interrupt based method to recognize overflow of Timer using necessary program. 6

2. Write a program to transmit a message serially "I want to score good in Microcontroller" to PC at the baud rate of 9600 bits/sec, using mode 1. 8

OR

(a) 1. Explain 8 bit Auto Reload & 16 bit mode for timer0. 6

2. Attempt all.

1. Calculate number to be loaded in TH1 for Baud rates of 9600, 4800, 2400 & 1200. 4

2. Explain following terms. 4

a. Synchronous & Asynchronous communication.

b. Full Duplex & Half Duplex mode of communication.

Q-6 (a) 1. Write a program which rotates stepper motor Clockwise if switch connected to P1.0 is pressed & stopped if switch connected to P1.1 is pressed. 6

2. Write a program which measures analog voltage on channel 0 of ADC0808 & display it on 16 x 2 LCD. Make suitable assumption & draw interfacing circuit. 8

OR

1. Write a program for DAC0808 which generates a Saw tooth signal on pressing switch connected to P1.0 & generates a 2 Volt signal on pressing switch connected to P1.1. 6

2. Write a program to display '8051' on seven segment led display with the help of interfacing diagram. 8